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Anticipation of novelty recruits reward system and hippocampus while promoting recollection

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The dopaminergic midbrain, which comprises the substantia nigra and ventral tegmental area (SN/VTA), plays a central role in reward processing. This region is also activated by novel stimuli, raising the possibility that novelty and reward have shared functional properties. It is currently unclear whether functional aspects of reward processing in the SN/VTA, namely, activation by unexpected rewards and cues that predict reward, also characterize novelty processing. To address this question, we conducted an fMRI experiment during which subjects viewed symbolic cues that predicted either novel or familiar images of scenes with 75% validity. We show that SN/VTA was activated by cues predicting novel images as well as by unexpected novel images that followed familiarity-predictive cues, an 'unexpected novelty' response. The hippocampus, a region implicated in detecting and encoding novel stimuli, showed an anticipatory novelty response but differed from the response profile of SN/VTA in responding at outcome to expected and 'unexpected' novelty. In a behavioral extension of the experiment, recollection increased relative to familiarity when comparing delayed recognition memory for anticipated novel stimuli with unexpected novel stimuli. These data reveal commonalities in SN/VTA responses to anticipating reward and anticipating novel stimuli. We suggest that this anticipatory response codes a motivational exploratory novelty signal that, together with anticipatory activation of the hippocampus, leads to enhanced encoding of novel events. In more general terms, the data suggest that dopaminergic processing of novelty might be important in driving exploration of new environments.
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Introduction

Single-neuron recordings in animals and recent functional magnetic resonance imaging (fMRI) studies in humans provide convergent evidence that the SN/VTA midbrain region is activated not only by reward (Schultz, 1998) but also by novel stimuli even in the absence of reinforcement (Schultz et al., 1997; Schott et al., 2004; Bunzeck and Duzel, 2006). SN/VTA activation by novelty raises the possibility that novelty might have intrinsic rewarding properties. If so, characteristics of reward processing, such as the temporal shift of responses in conditioning, should also hold for novelty processing. In reward anticipation paradigms, dopaminergic neurons code reward prediction when the contingency between a predictive stimulus and subsequent reward delivery has been learned. Specifically, these neurons respond to the first reliable predictor of reward but no longer to receipt of reward (Ljungberg et al., 1992; Schultz et al., 1992, 1997; Schultz, 1998). Whether novelty processing in the SN/VTA also shows these reward-related properties is unclear.

The hippocampus is critical in formation of episodic long-term memories for novel events (Vargha-Khadem et al., 1997; Duzel et al., 2001) and believed to provide the major input for a novelty signal in SN/VTA (Lisman and Grace, 2005). Dopamine released by SN/VTA neurons, in turn, is critical for stabilizing and maintaining long-term potentiation (LTP) and long-term depression (LTD) in hippocampal region CA1 (Frey et al., 1990, 1991; Huang and Kandel, 1995; Sajikumar and Frey, 2004; Lemon and Manahan-Vaughan, 2006; for a review see Jay, 2003). fMRI data have shown that joint SN/VTA and hippocampal activation is associated with successful long-term memory formation (Schott et al., 2006) and reward-related improvement in novel stimulus encoding (Wittmann et al., 2005; Adcock et al., 2006). In light of such converging evidence, recent models of hippocampus-dependent memory formation emphasize a functional relationship between novelty detection in the hippocampus and enhancement of hippocampal plasticity by novelty-induced dopaminergic modulation arising from the SN/VTA (Lisman and Grace, 2005). Therefore, the question whether the SN/VTA is activated by anticipating novelty goes beyond a conceptual understanding of the relationship between novelty and reward to embrace mechanisms of hippocampal plasticity. Furthermore, it has recently been suggested that understanding the relationship between novelty and reward-processing in SN/VTA might reveal links between motivation, novelty-seeking

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Exploring a novel environment improves motivation and promotes recall of words

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Active exploration of novel environments is known to increase plasticity in animals, promoting long-term potentiation in the hippocampus and enhancing memory formation. These effects can occur during as well as after exploration. In humans novelty's effects on memory have been investigated with other methods, but never in an active exploration paradigm. We therefore investigated whether active spatial exploration of a novel compared to a previously familiarized virtual environment promotes performance on an unrelated word learning task. Exploration of the novel environment enhanced recall, generally thought to be hippocampus-dependent, but not recognition, believed to rely less on the hippocampus. Recall was better for participants that gave higher presence ratings for their experience in the virtual environment. These ratings were higher for the novel compared to the familiar virtual environment, suggesting that novelty increased attention for the virtual rather than real environment; however, this did not explain the effect of novelty on recall.

Keywords: novelty, exploration, memory enhancement, episodic memory, dopaminergic mesolimbic system

INTRODUCTION

Animal studies have consistently shown that exploration of a novel environment can promote long-term potentiation (LTP) in the hippocampus and specifically in the dentate gyrus, thereby improving memory encoding (Li et al., 2003; Straube et al., 2003a; Davis et al., 2004; Sajikumar and Frey, 2004; Sierra-Mercado et al., 2008). The modulatory effect of novelty on learning has been suggested to rely on the functional connections between the hippocampus and the substantia nigra/ventral tegmental area (SN/VTA), components of the mesolimbic dopaminergic system, by regulating the entry of new information into long-term memory (Lisman and Grace, 2005). When novelty is detected, a novelty signal from the hippocampus activates dopaminergic neurons in the SN/VTA. This results in dopamine being released in the hippocampus, where dopamine increases plasticity by enhancing LTP. Interestingly, increased plasticity not only occurs during exploration of the novel environment, but lasts up to 10 min afterwards (Li et al., 2003; Straube et al., 2003a; Kentros et al., 2004).

Direct evidence for a link between exploration of novel environments and increased plasticity comes from animal studies, but also in humans evidence for such a link has been found. Active as compared to passive exploration of a virtual environment (VE) has been shown to improve recall of allocentric spatial information (Voss et al., 2011; Plancher et al., 2012, 2013). Memory improvements have been observed for familiar items when these were presented in the context of novel scene stimuli (Bunzeck and Düzel, 2006). Similarly, passive exposure to novel scenes has been shown to improve recall of words presented in a task after exposure (Fenker et al., 2008). In this latter study either novel or previously familiarized scenes were presented for which indoor/outdoor judgments had to be made. After exposure to novel compared to familiar scenes recall and recollection on an unrelated word learning task were superior, whereas no such improvements were observed for familiarity-based recognition (Fenker et al., 2008). Also memory improvements have been found for items that stand out in the to-be-learned list, a benefit that may be novelty-related. For example, in the Von Restorff or distinctiveness effect memory formation is better for words presented in distinctive novel fonts than for words presented in a standard font (Von Restorff, 1933; Bruce and Gaines, 1976; Schmidt, 1985; Hunt, 1995; Dunlosky et al., 2000; Geraci and Rajaram, 2004; Geraci and Manzano, 2010; Rangel-Gomez and Meeter, 2013). Furthermore, the anticipation of a novel stimulus has been shown to enhance memory encoding for unexpected novel events (Wittmann et al., 2007). However, no studies so far have looked at whether active exploration of a novel environment promotes memory relative to familiar environments, or whether the effects of exploration extend to a subsequent learning task in humans.

In the present study, participants performed a word learning task either after exploring a previously familiarized or a novel VE from a first-person perspective in a within-subjects design (see **Figure 1**). In line with the findings above, we expected learning to be improved after exploring a novel VE, relative to a familiarized one. As was found by Fenker et al. (2008) and given that exploration of spatial novelty has been strongly associated with the hippocampus (Lisman and Grace, 2005; Jeewajee et al., 2008; Bast et al., 2009; Kaplan et al., 2012), we expected a learning benefit for free recall, which is thought to depend on the hippocampus (Yonelinas et al., 2002), and not for recognition, believed to rely mainly on the perirhinal cortex (Aggleton and Brown, 1999; Wan et al., 1999; Duzel et al., 2001; Fernandez et al., 2002; Yonelinas et al., 2002).

MATERIALS AND METHODS

PARTICIPANTS

Thirty-two participants naïve to the aims of the study participated in return for 15€ or course credits. Two quit the experiment

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in retrospective

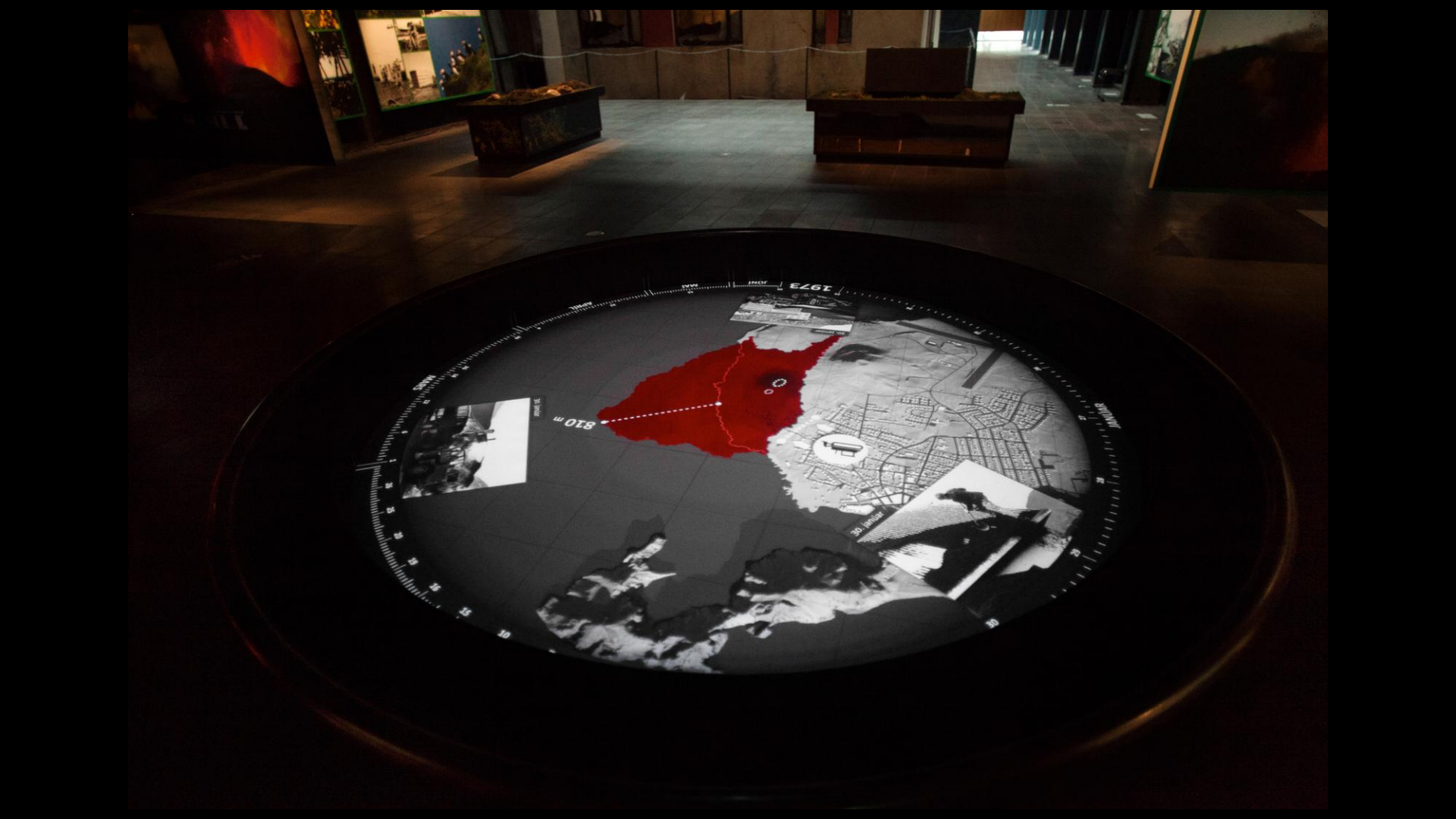

Uppgröftur
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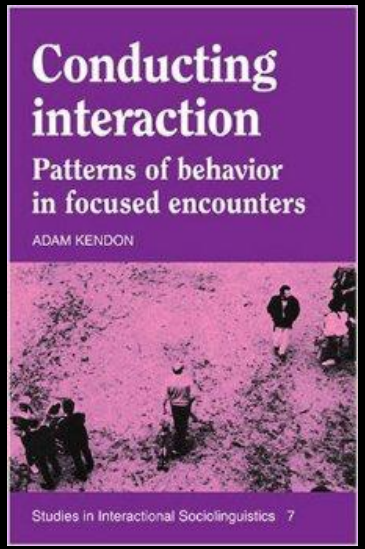


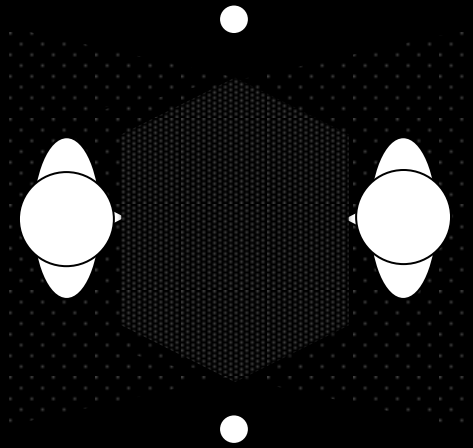


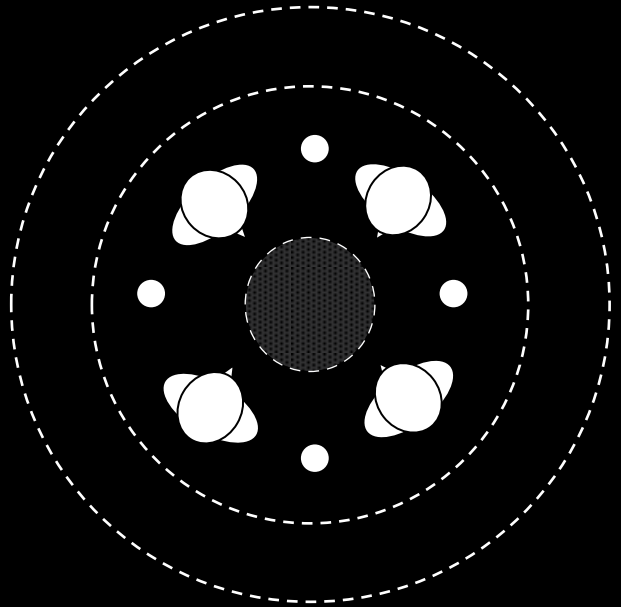

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Conducting interaction
Patterns of behavior in focused encounters
ADAM KENDON
Studies in Interactional Sociolinguistics 7






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